Data_Analyst — Step 5

Build Data Analysis Projects

Detailed Learning Notes • CareerCompass

1. Learning Objective

Apply Excel, SQL, Python and BI skills to complete, business-oriented analytical projects.

2. Core Concepts — Detailed Breakdown

Project framing

Start with a stakeholder question and define the decision the analysis should support.

Data pipeline

Document source, grain, cleaning, transformations and metric definitions.

Analysis

Use SQL/Excel/Python to validate, aggregate and investigate the data.

Dashboard/report

Present a small number of decision-oriented views with clear definitions and filters.

Recommendations

Recommendations should follow from evidence and include assumptions and limitations.

3. Recommended Learning Workflow

- Write the stakeholder question.
- Create a metric dictionary.
- Validate raw data.
- Analyze with SQL/Python.
- Build dashboard.
- Write recommendations and limitations.

5. Hands-on Project / Practice

Build a customer-sales or marketing-performance project. Include SQL extraction, Python cleaning/EDA, a Power BI or Tableau dashboard and a one-page executive summary with 3–5 recommendations.

6. Common Mistakes & Pitfalls

- No stakeholder question.
- Dashboard with too many unrelated charts.
- Recommendations not supported by analysis.
- No validation of totals.
- Using sample/tutorial data without understanding it.

7. Completion Checklist

• My project has a clear decision question.

• Metrics are defined.

• Results are reproducible.

• Dashboard is readable.

• Recommendations are evidence-based.

8. Interview / Assessment Questions

• How did you validate your numbers?

• What insight changed your recommendation?

• Why did you choose this chart?

• What would you investigate next?

• What assumptions limit your conclusion?

9. Recommended References

Data Analyst Bootcamp: https://www.youtube.com/watch?v=PSNXoAs2FtQ

Microsoft Power BI: https://learn.microsoft.com/en-us/training/paths/prepare-visualize-data-power-bi/

Tableau Training: https://www.tableau.com/learn/training

Data Analysis with Python: https://www.youtube.com/watch?v=GPVsHOlRBBI

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.